

Placebo Tests and Negative Controls

Experimentation, A/B Testing and Causal Inference (SDA-DSC-213), SDAIA Academy. Handout for Day 4, Hour 2 (slide 137). Same story as Day 3: a tutoring programme at one school.

Step 1. The idea

Sensitivity analysis asks how bad a hidden confounder **could** be. Falsification tests look for evidence that one **is** there. The trick: run the same analysis where the true effect must be zero. If you find an effect anyway, the design is leaking.

Step 2. Three checks on the tutoring estimate

Check	What we run	Effect must be	We found	Verdict
Negative-control outcome	same adjustment for X, but the outcome is the art grade , which tutoring did not cover	0	+0.4 (CI –2.1 to 2.9)	clean
Negative-control exposure	students who signed up but never attended versus never signed up, adjusted for X	0	+5.6 (CI 2.8 to 8.4)	leak
Placebo timing	DiD for the library, pretending it opened a year early (handout 3)	0	0	clean

Step 3. Reading the leak

Signing up without attending cannot teach anyone anything, yet it comes with +5.6 points after adjusting for prior grade. Something that makes students sign up also makes them score higher, and prior grade does not capture it. That is the hidden U from the sensitivity handout, seen in the data.

estimate +9 = tutoring effect + leak
leak is at least about 5.6 → the tutoring effect is closer to +3 than to +9

This is the same step as Lab 3's negative-control outcome check: the same adjustment set applied to an outcome that cannot respond.

Step 4. Which check to use

Design	Natural falsification test
Adjustment (g-formula, IPW, matching)	negative-control outcome; negative-control exposure
Difference-in-differences	placebo timing (pre-period); placebo unit (an untreated school)

Regression discontinuity	covariate continuity at the cutoff; placebo cutoffs at 65 and 75
Instrumental variables	the instrument should not predict pre-treatment outcomes
Experiment	A/A test; balance on pre-treatment covariates

Three things to remember

1. A falsification test can only fail you; passing it supports the design, it does not prove it.
2. Choose a negative control that shares the confounding path but not the causal one.
3. Report the passes and the fails together with the estimate.